

Short-Channel Effects in Nanoscale CMOS

KKKC6353 Nanoelectronic Devices · Lecture 1, Part 3 of 3

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One problem, not a list

1

The usual framing

Short-channel effects taught as a **list** of five or nine items to memorise: CLM, DIBL, punchthrough, velocity saturation, hot carriers…

2

One problem

The gate should be the only electrode controlling the source–drain **barrier**. In a short channel, **the drain competes**. Everything else is a symptom, or a consequence of high fields.

3

One family of solutions

Halos, SOI, FinFET, GAA are the **same move**: shrink the volume the drain can reach, so the gate wins. The rest of the module starts where that move runs out.

1

One problem: electrostatic integrity

When is a channel 'short'?

The natural length λ

- Solve Poisson's equation in two dimensions: the drain's influence decays over the **natural length λ**
- The perturbation falls off as **exp(−L/2λ)**
- **Thinner body** and **electrically thinner dielectric** both shrink λ, but only as square roots
- **More gates** change the prefactor: the strongest lever

$$\lambda = \sqrt{\frac{\varepsilon_{Si}}{\varepsilon_{ox}} t_{Si}\, t_{ox}} \qquad L \gtrsim 5\lambda \text{ to } 10\lambda$$

The design rule used throughout the industry.

$$\lambda_{single} = \sqrt{\frac{\varepsilon_{Si} t_{Si} t_{ox}}{\varepsilon_{ox}}} \quad \lambda_{double} = \sqrt{\frac{\varepsilon_{Si} t_{Si} t_{ox}}{2\varepsilon_{ox}}} \quad \lambda_{GAA} \approx \sqrt{\frac{\varepsilon_{Si} t_{Si} t_{ox}}{4\varepsilon_{ox}}}$$

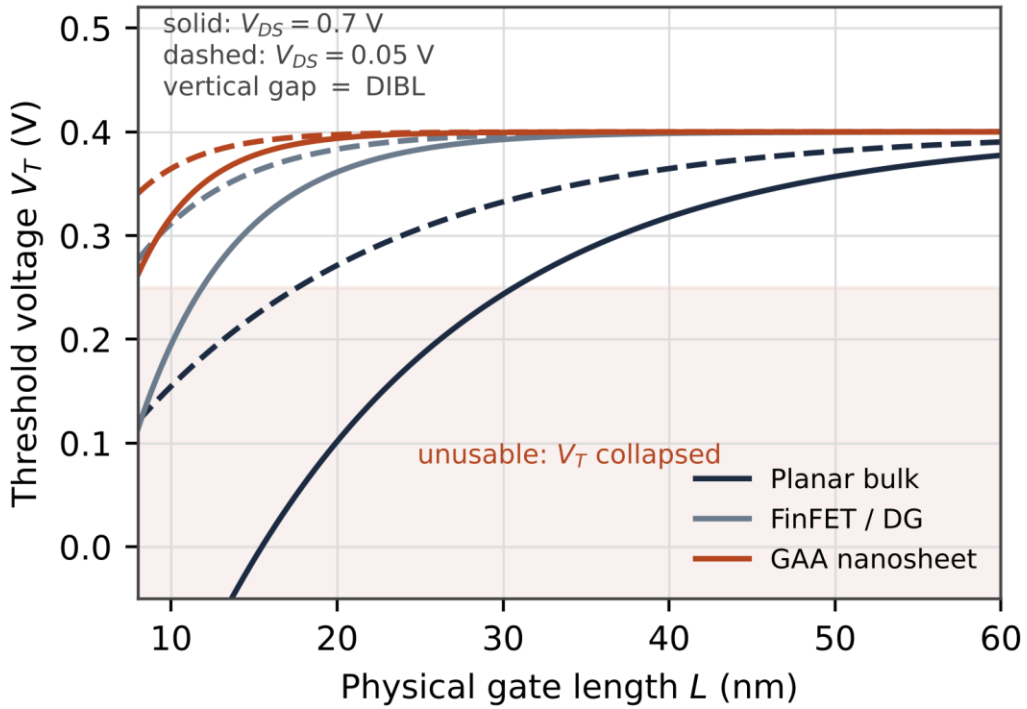
Single gate, double gate (FinFET), gate-all-around.

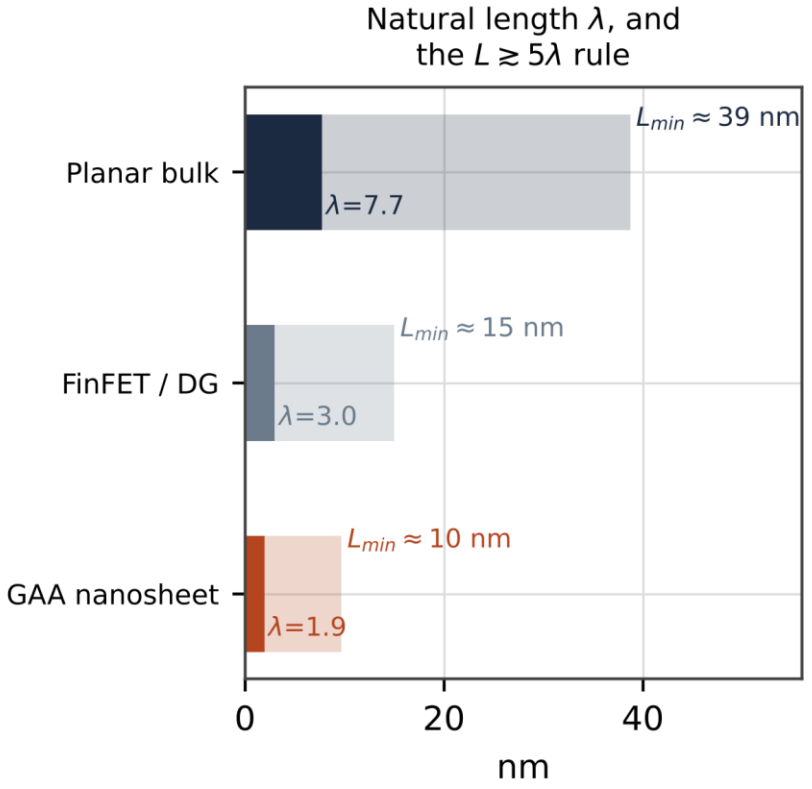
Putting numbers in (EOT = 1 nm)

Architecture	Body	λ	$L_{min} \approx 5\lambda$
Planar bulk	x_{dm} = 20 nm	7.8 nm	≈ 39 nm
FinFET / double gate	t_{Si} = 6 nm	3.0 nm	≈ 15 nm
GAA nanosheet	t_{Si} = 5 nm	1.9 nm	≈ 10 nm

Compare with history: planar bulk was pushed to **35 nm gates** (Intel 65 nm, 2005) with halos and thin oxides; FinFETs carried 22 nm to 3 nm; the 2022 IRDS projects GAA gate length to **saturate near 12 nm**. One line of algebra gets the order and scale of every transition.

Roll-off and DIBL for three architectures





$V_T(L) = V_T^{\infty} - (a + bV_{DS})\exp(-L/2\lambda)$, same a and b, only λ changed.

The gap between solid and dashed curves is **DIBL**. Each architecture runs out at about 5λ.

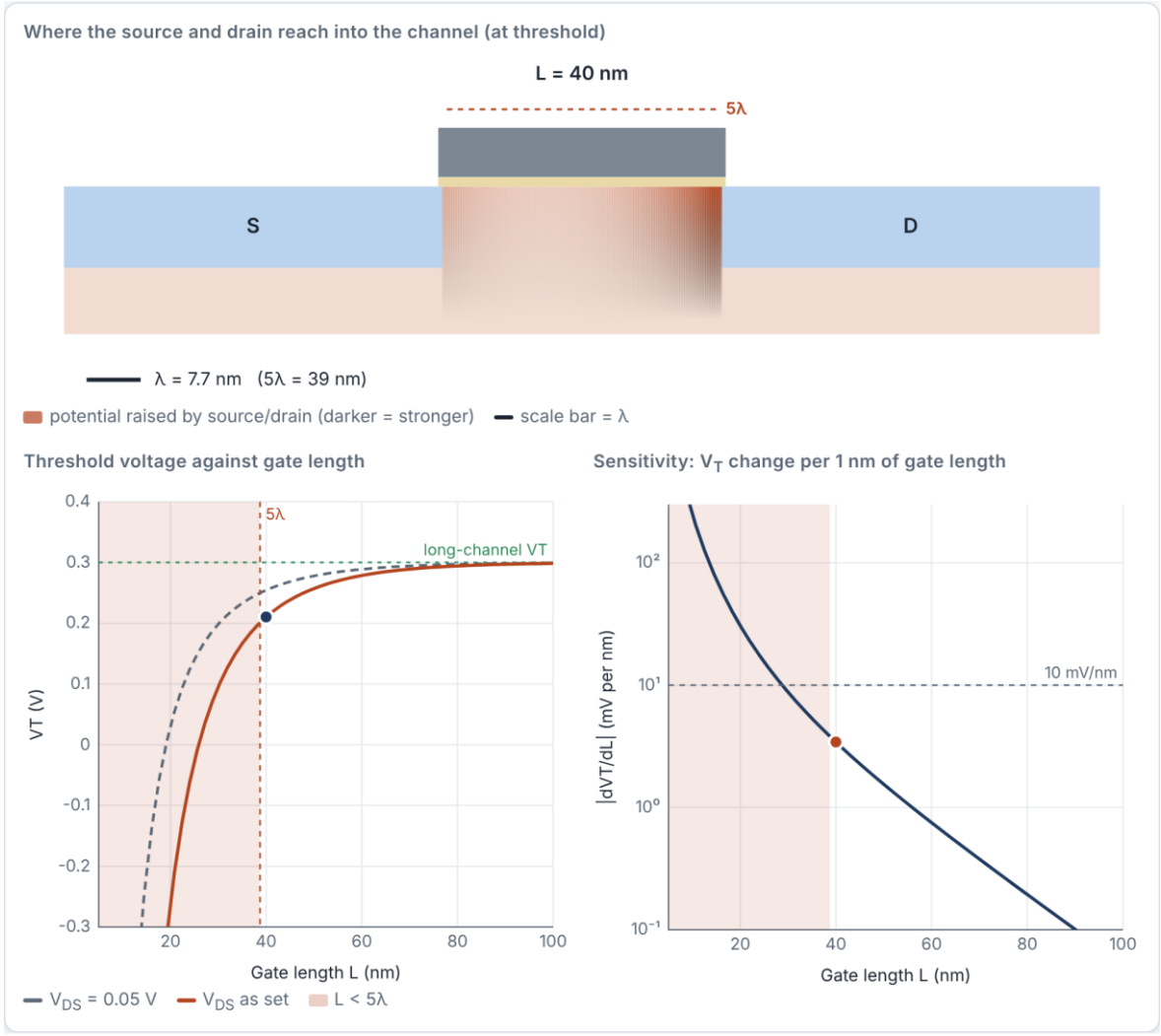
Animation: natural length and threshold roll-off

Interactive animation 2

Natural length and threshold roll-off

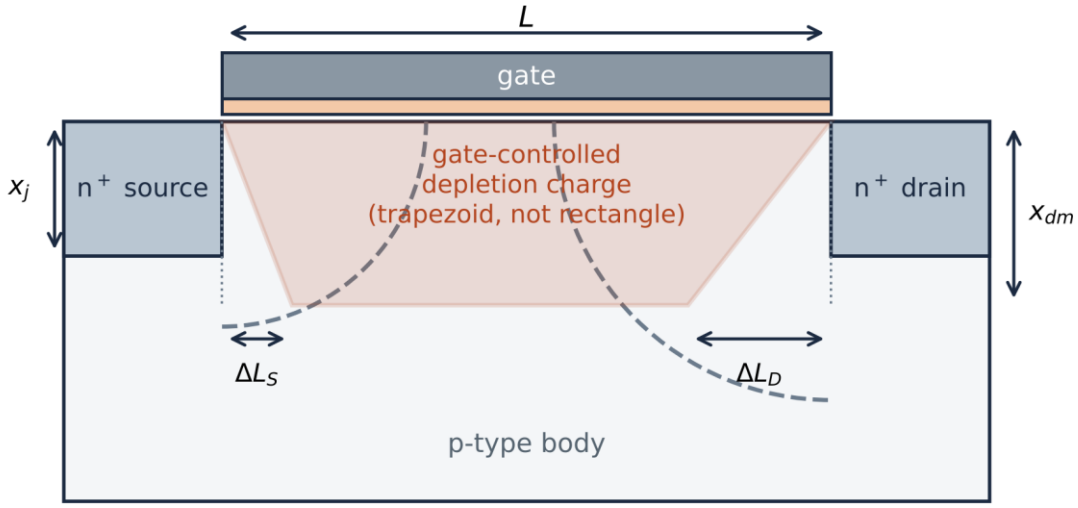
faizverse.github.io/nanoelectronic-devices/L1/vt-rolloff.html

- Bulk: slide L from 100 nm to 20 nm and watch the source and drain zones merge
- Read the V_T sensitivity at 30 nm: mV per nm of gate length
- Thin the oxide, then the body: square-root levers
- Switch to FinFET and GAA: the lever the industry used



The classical picture: charge sharing

- Long-channel V_T assumed the gate supports **all** the depletion charge
- In a short device the junctions support the corners: the gate's share is a **trapezoid**, so V_T falls ($\Delta V_T \propto x_j/L$)
- **Use as intuition, not a model:** a 1-D story about a 2-D problem
- No drain-voltage dependence, so it **cannot explain DIBL**; underestimates roll-off below ~50 nm



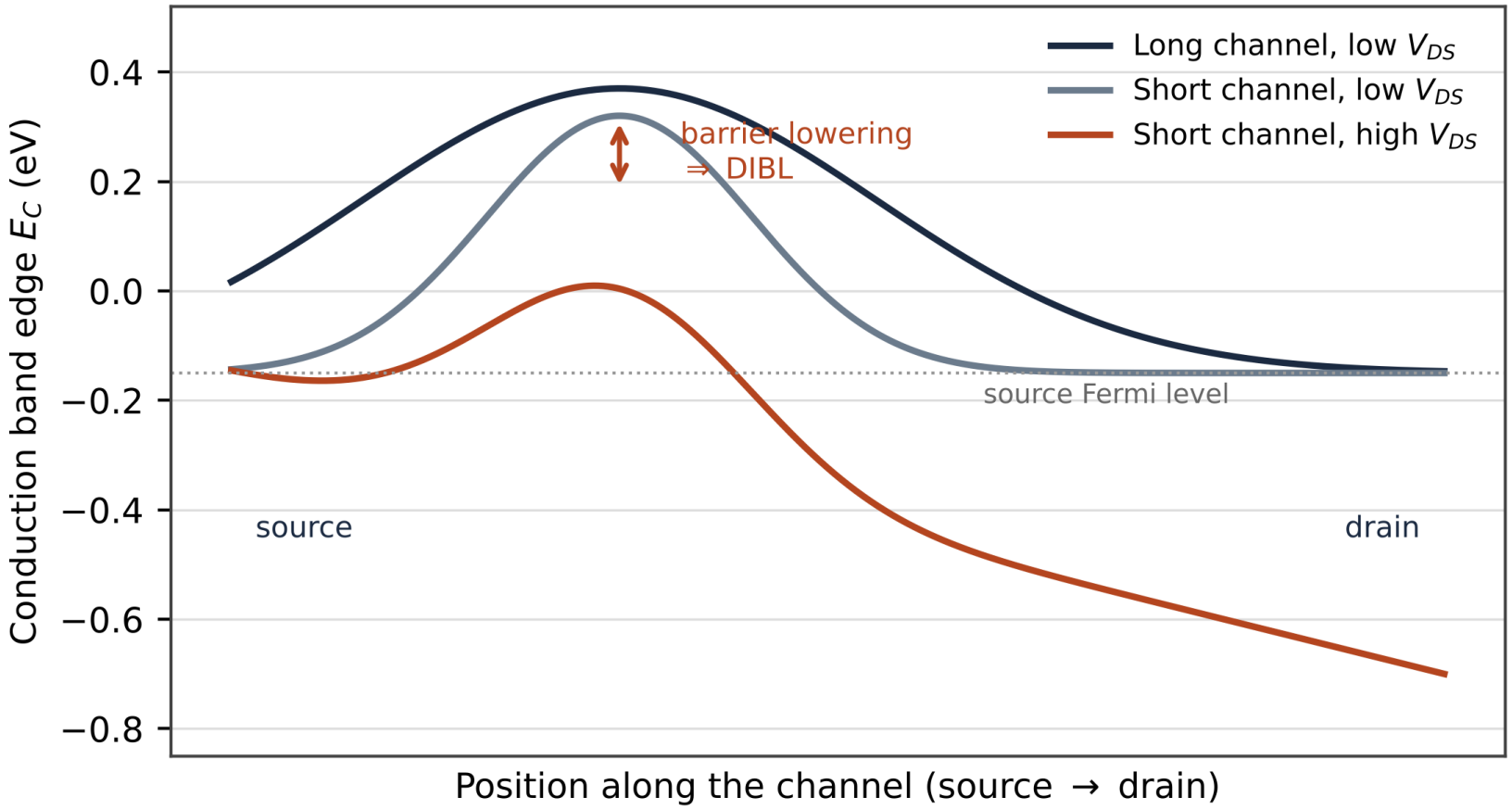
$$\Delta L_{S,D} \approx x_j\left(\sqrt{1+\frac{2x_{dS,dD}}{x_j}}-1\right)$$

2

Symptoms: the gate loses control

Read the barrier figure carefully: the next five symptoms are all in it.

The barrier the gate is supposed to control



Long channel: high, wide barrier set by the gate. Short channel: lower even at low V_{DS} (**roll-off**). Raise V_{DS}: lower still (**DIBL**). Barrier gone: **punchthrough**.

Threshold-voltage roll-off

What happens

V_T falls as L falls. The junctions support part of the depletion charge, or equivalently the source and drain raise the channel potential, so less gate voltage is needed to invert.

Why it matters

Not that V_T is lower (that can be compensated) but that V_T becomes a **steep function of L**. A **1 nm** lithographic variation becomes **tens of mV** of V_T spread (Section 5).

Drain-induced barrier lowering (DIBL)

- Raise V_{DS}: the barrier falls further, so **V_T falls with drain voltage**
- Well scaled: **20–60 mV/V**; above **100 mV/V**, poor control
- The single number engineers measure for electrostatic integrity
- Costs: higher off-current at high V_{DS}, and higher output conductance (lost analogue gain)
- **Pitfall:** 'eliminated by scaling junctions and raising doping' was true at 0.25 µm, not at 15 nm

$$\mathrm{DIBL} = -\frac{\Delta V_T}{\Delta V_{DS}} = \frac{V_T|_{low} - V_T|_{high}}{V_{DS,\,high} - V_{DS,\,low}} \quad [\mathrm{mV/V}]$$

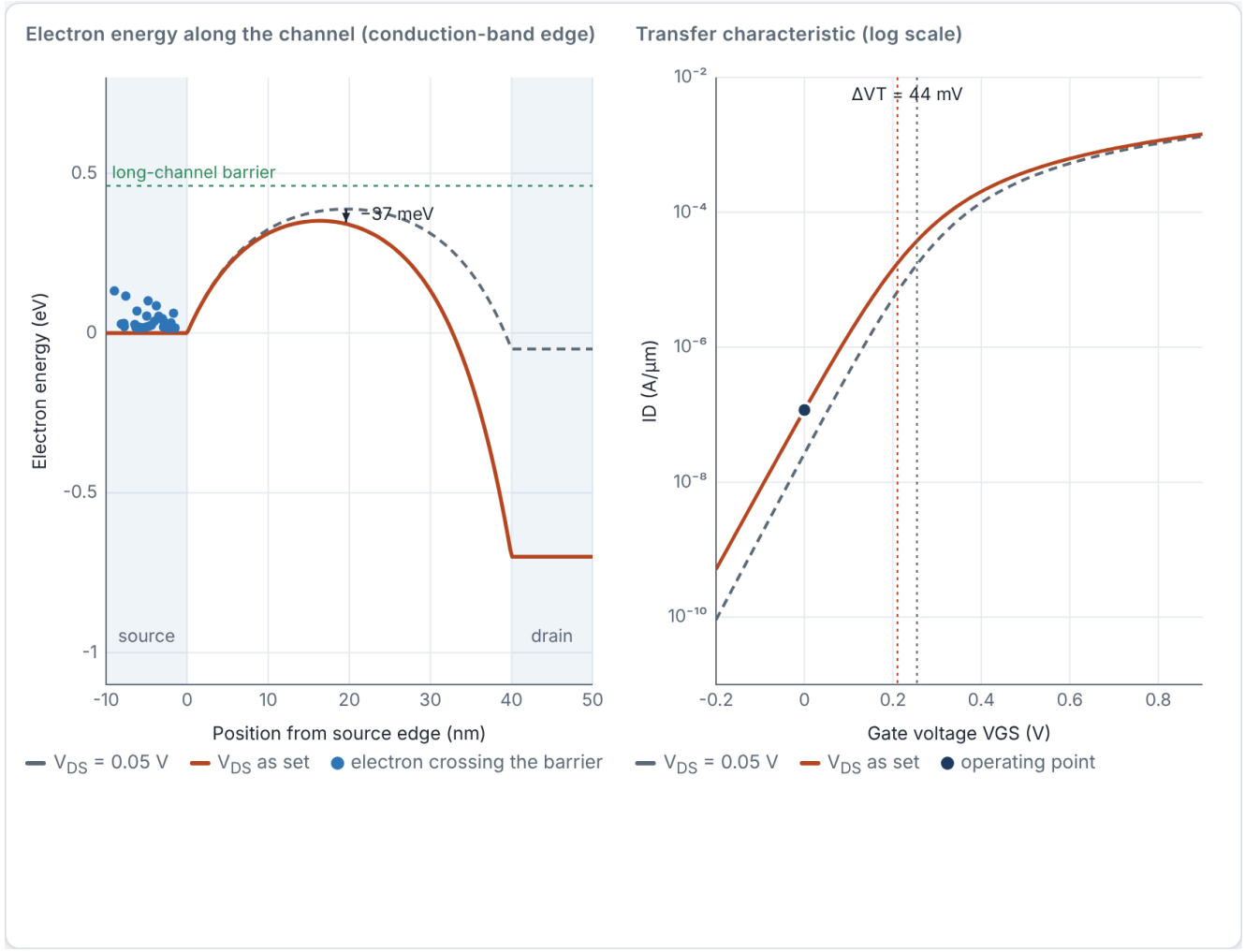
Animation: drain-induced barrier lowering

Interactive animation 3

Drain-induced barrier lowering

faizverse.github.io/nanoelectronic-devices/L1/dibl.html

- Bulk, L = 100 nm: sweep V_{DS}; the barrier top does not move
- L ≈ 40 nm (5λ): DIBL ≈ 70 mV/V
- L = 25 nm: electrons pour over the lowered barrier
- Switch to FinFET and GAA at the same L



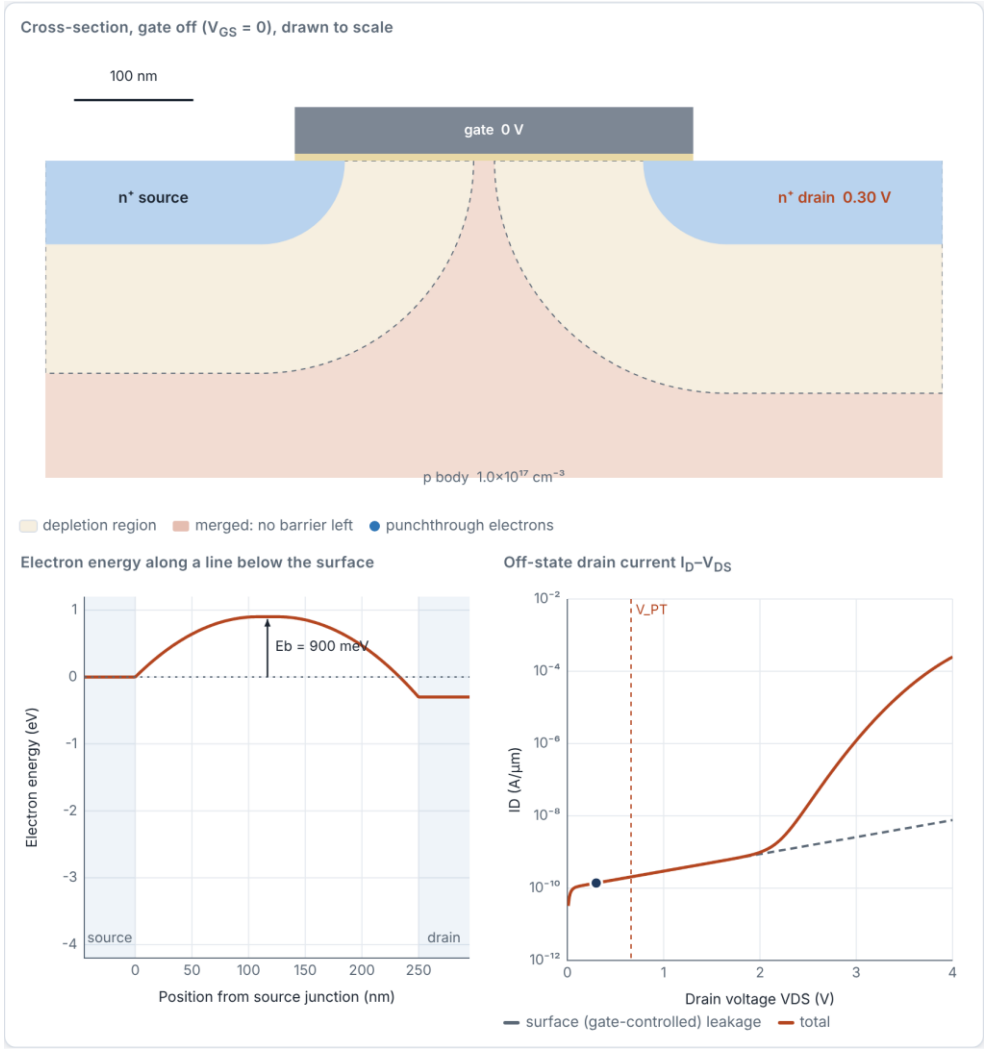
Animation: punchthrough

Interactive animation 4

Punchthrough

faizverse.github.io/nanoelectronic-devices/L1/punchthrough.html

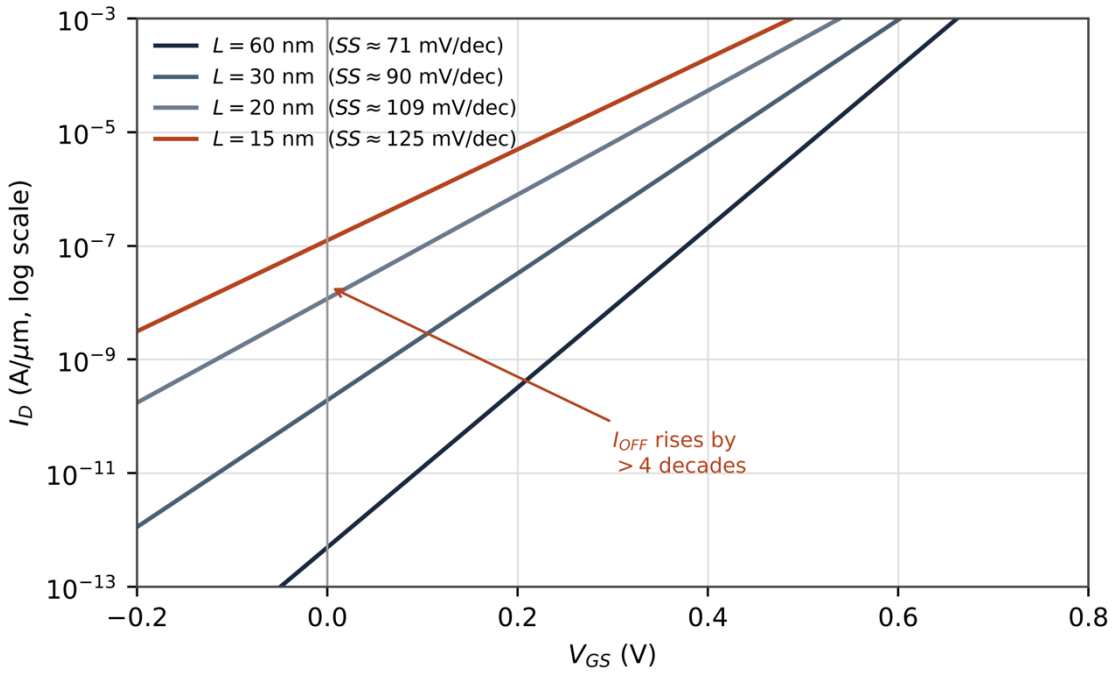
- Source and drain depletion regions **merge below the surface**; the barrier vanishes where the gate cannot reach
- Current then rises almost uncontrollably with V_{DS}
- The terminal case of lost control, not a separate effect
- Suppressed by higher body doping, **halo/retrograde implants**, or by removing the deep body (SOI, multi-gate)



Subthreshold swing degradation

- The drain adds its own coupling, $\mathbf{C_{drain}}$, to the capacitive divider
- Swing rises above the long-channel value; **100–120 mV/dec** in a badly scaled device
- Three symptoms **compound**: curves shift left (roll-off, DIBL) and flatten (swing)
- Off-current at $V_{GS} = 0$ rises by **over four decades**

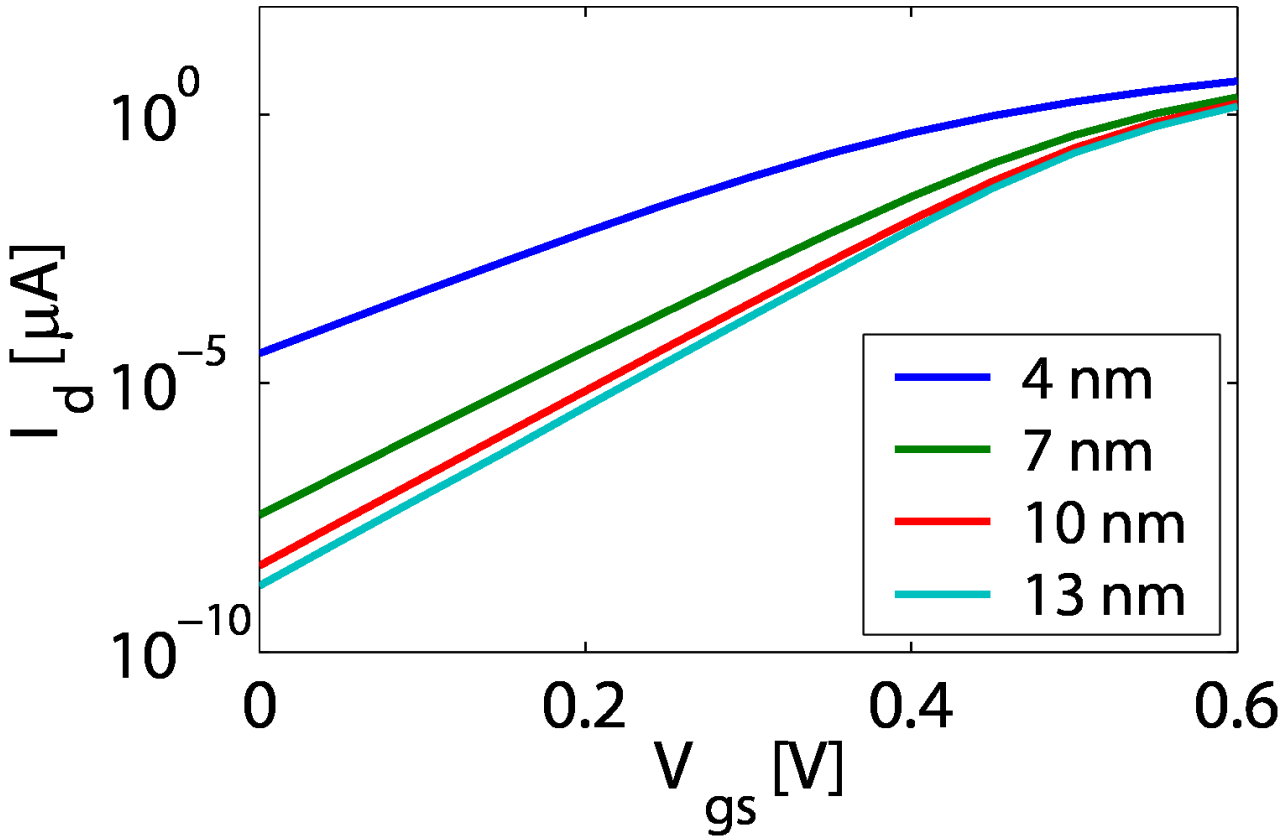
$$SS \approx \left(1 + \frac{C_d + C_{drain}}{C_{ox}}\right)\frac{kT}{q}\ln 10$$



Planar bulk at $V_{DS} = 0.7$ V as L shrinks.

The same trends from first principles

- Full quantum-transport simulation of an ultra-scaled transistor
- Gate length falls from **13 nm to 4 nm**
- Off-current rises by orders of magnitude; the swing flattens
- **Not independent effects:** they multiply



Quantum-transport simulation, gate length 13 nm to 4 nm. Reproduced from ETH Zürich lecture slides.

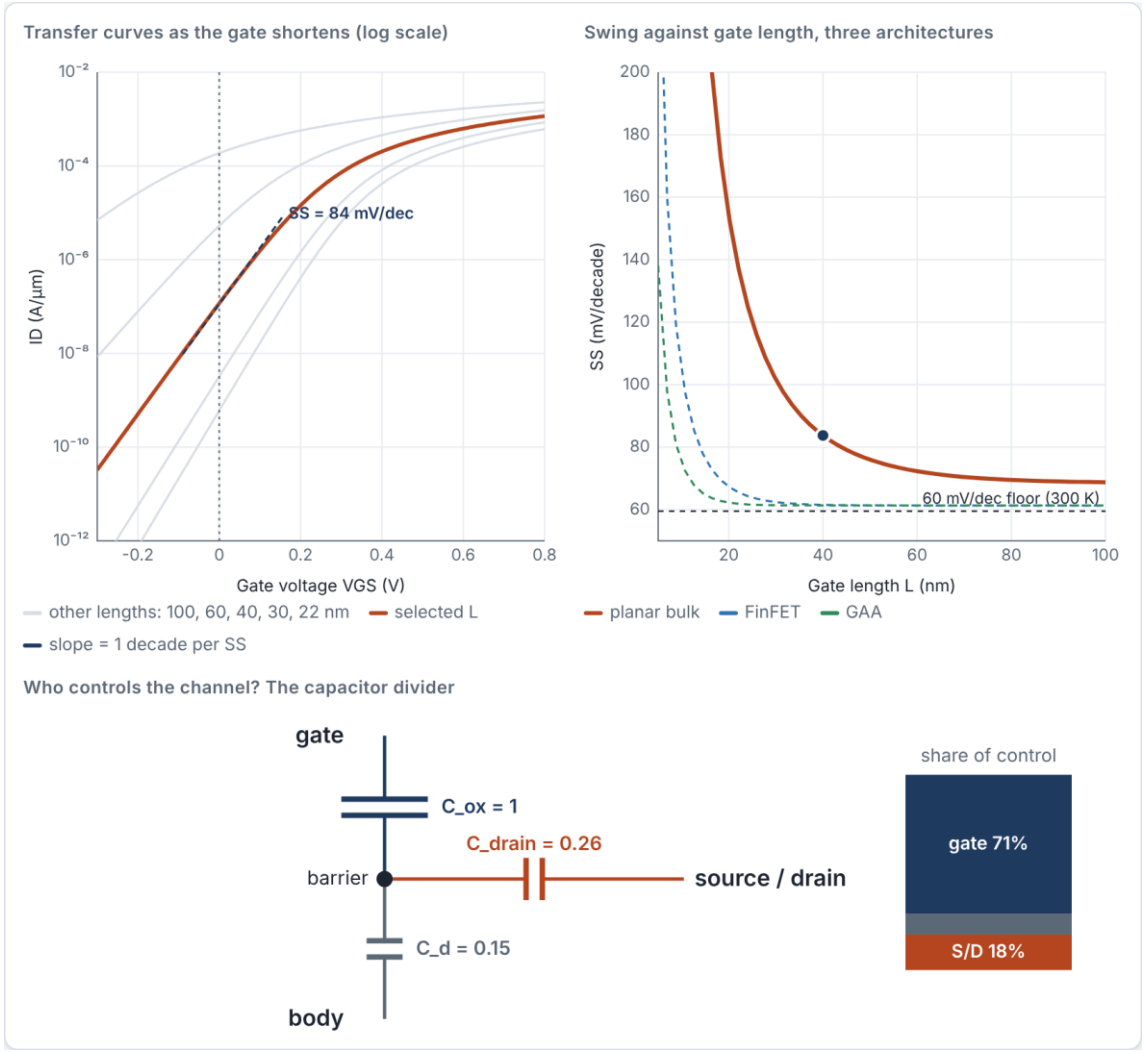
Animation: subthreshold swing degradation

Interactive animation 5

Subthreshold swing degradation

faizverse.github.io/nanoelectronic-devices/L1/subthreshold-swing.html

- Bulk, L = 100 nm: SS ≈ 69 mV/dec
- Slide to 25 nm: the drain capacitor grows; SS passes 100 mV/dec
- FinFET and GAA: C_d vanishes, C_{drain} shrinks
- No setting beats 59.5 mV/dec: why?



Channel-length modulation

- Past saturation the pinch-off point moves toward the source: **effective L shortens**
- Current keeps rising instead of saturating
- Mostly an **analogue** concern: sets the intrinsic gain $g_m r_o$
- In digital logic it is secondary to DIBL
- **Notation:** λ_{CLM} is the circuit parameter, unrelated to the natural length λ

$$I_D \approx I_{D,\,sat}\,(1 + \lambda_{CLM} V_{DS})$$

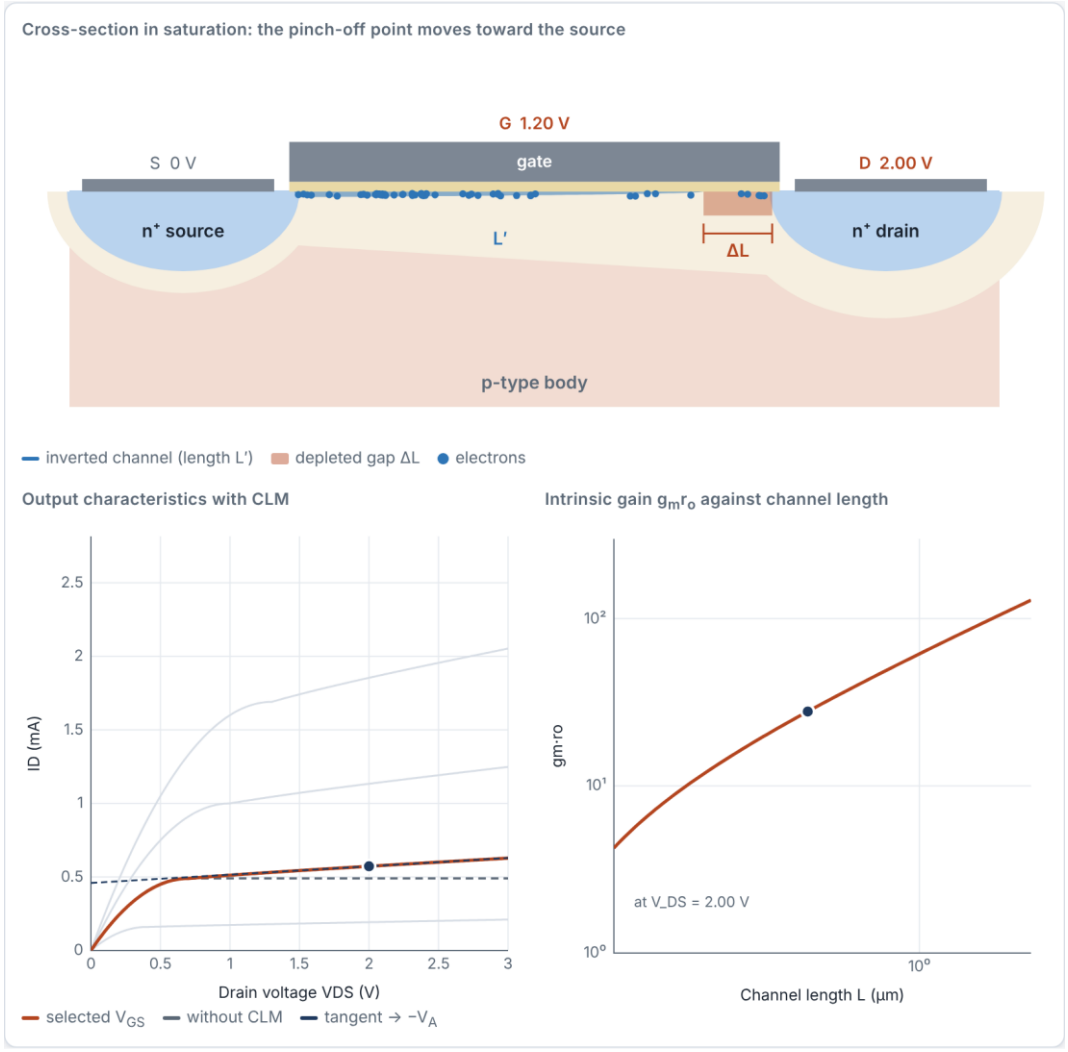
Animation: channel-length modulation

Interactive animation 6

Channel-length modulation

faizverse.github.io/nanoelectronic-devices/L1/channel-length-modulation.html

- L = 1 µm, V_{GS} = 1.2 V: sweep V_{DS} and watch the pinch-off point leave the drain
- Same ΔL at L = 0.25 µm: a much larger fraction
- Read the Early voltage and intrinsic gain



3

Symptoms: high fields

0.7 V across 15 nm is 4.7×10^5 V/cm on average, with peaks several times higher.

Velocity saturation

- In a short channel the device is in velocity saturation over most of its range
- **Linear in overdrive, not quadratic, and independent of L**
- Drive current stops improving when you shorten the channel
- $g_m \approx W C_{ox} v_{sat}$ becomes a geometric constant
- Connects to quasi-ballistic injection (Part 2, Section 4.1; Animation 7)

$$I_{D,\,sat} \approx W C_{ox} v_{sat} (V_{GS} - V_T)$$

Mobility degradation and impact ionisation

Mobility at the surface

Vertical field ~10^6 V/cm presses carriers against the interface: μ_{eff} falls to **200–400** against bulk >1400 $cm^2/V{\cdot}s$ (Part 2, Section 4.3).

Impact ionisation

A carrier with more than **~1.5 eV** creates an electron–hole pair. Holes flow to the body: **substrate current**, peaking near $V_{GS} \approx V_{DS}/2$. Can trigger the parasitic bipolar.

Mostly historical today

Carriers cannot gain 1.5 eV from a 0.7 V supply. Still matters in I/O and analogue blocks, and as the **floating-body effect** in SOI and FinFETs.

Hot carriers, and what limits reliability now

Hot carriers

Above the Si/SiO_2 barrier (**3.1 eV** electrons, 4.8 eV holes) a carrier can be injected and trapped, shifting V_T. Why the LDD was invented. Now driven by many moderate-energy carriers.

BTI

Bias-temperature instability (NBTI, PBTI): V_T shift under gate bias at temperature. Usually the **leading** wear-out mechanism today.

TDDB

Time-dependent dielectric breakdown: slow degradation of the gate dielectric under field.

Electromigration

In the interconnect, not the transistor, but often the actual lifetime limiter of a product.

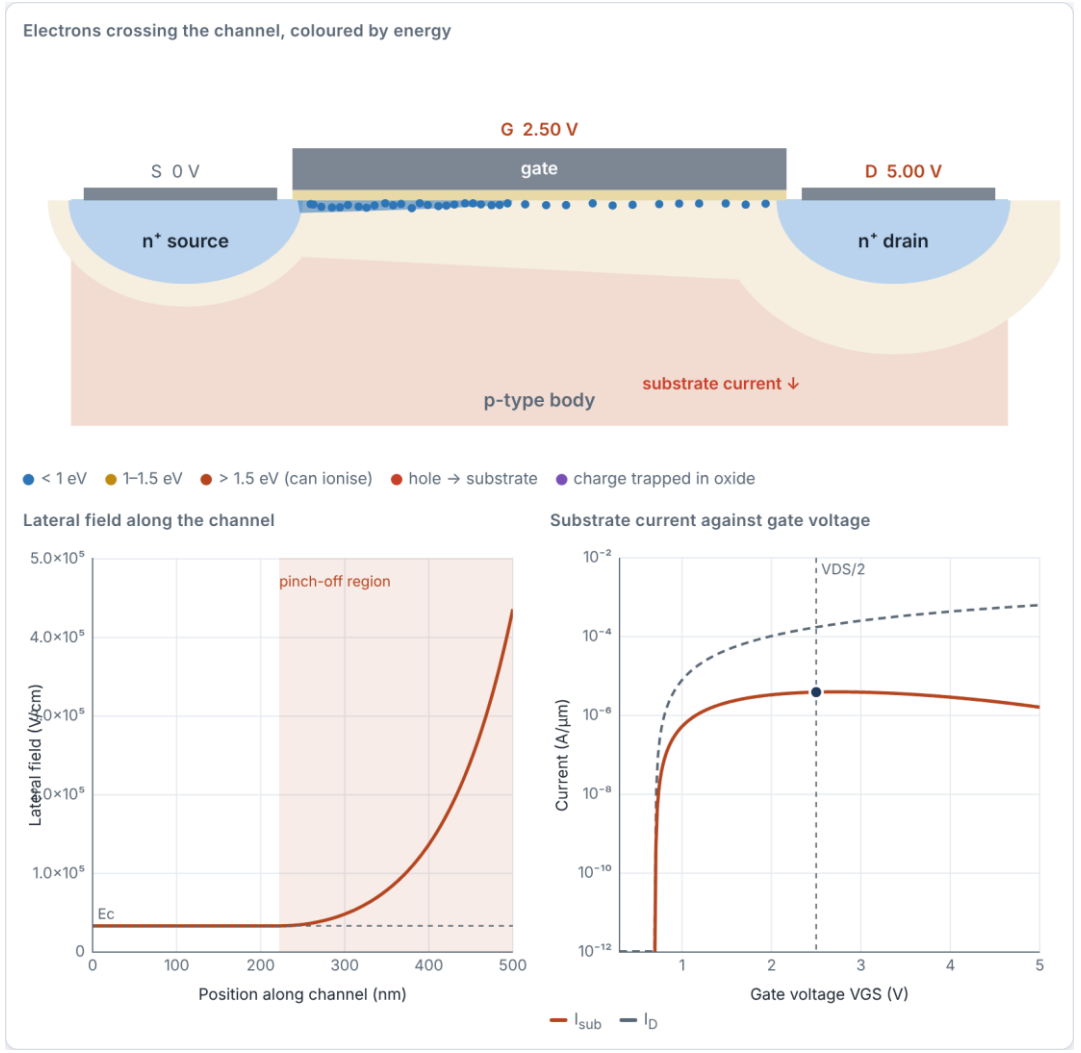
Animation: hot carriers and impact ionisation

Interactive animation 8

Hot carriers and impact ionisation

faizverse.github.io/nanoelectronic-devices/L1/hot-carriers.html

- 1990 · 5 V preset: electrons turn hot near the drain; holes drop into the substrate
- Find the I_{sub} peak against V_{GS}
- Turn on the LDD: the field peak flattens
- Today · 0.7 V: nothing gets hot. Energy, not field, is the limit



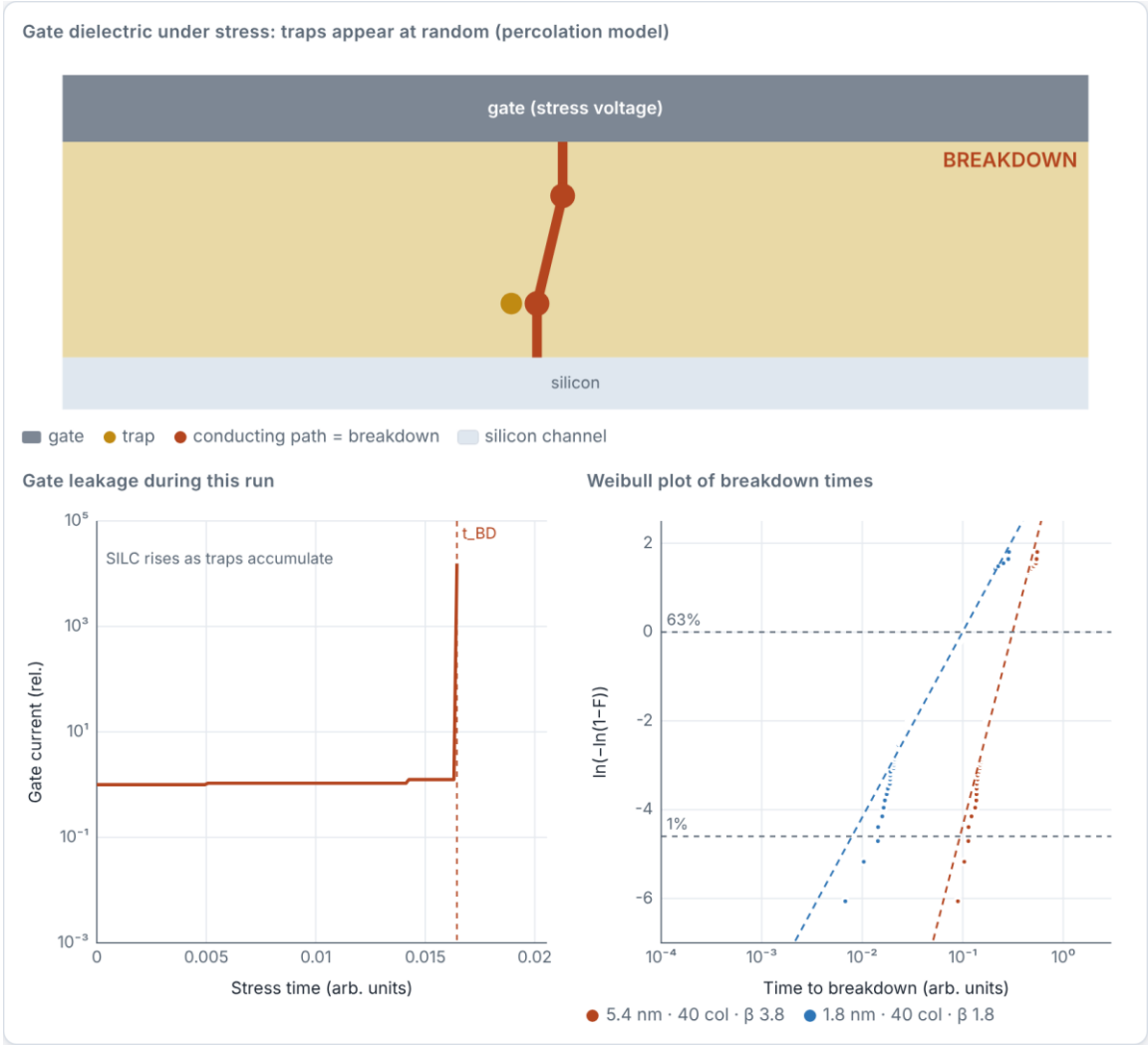
Animation: oxide breakdown (TDDB)

Interactive animation 9

Oxide breakdown (TDDB)

faizverse.github.io/nanoelectronic-devices/L1/oxide-breakdown.html

- Stress one device: traps appear until a path links gate to channel
- Run again: a different breakdown time
- Test 300 devices at 5.4 nm and at 1.8 nm: thin oxides fail earlier and spread wider (smaller Weibull β)

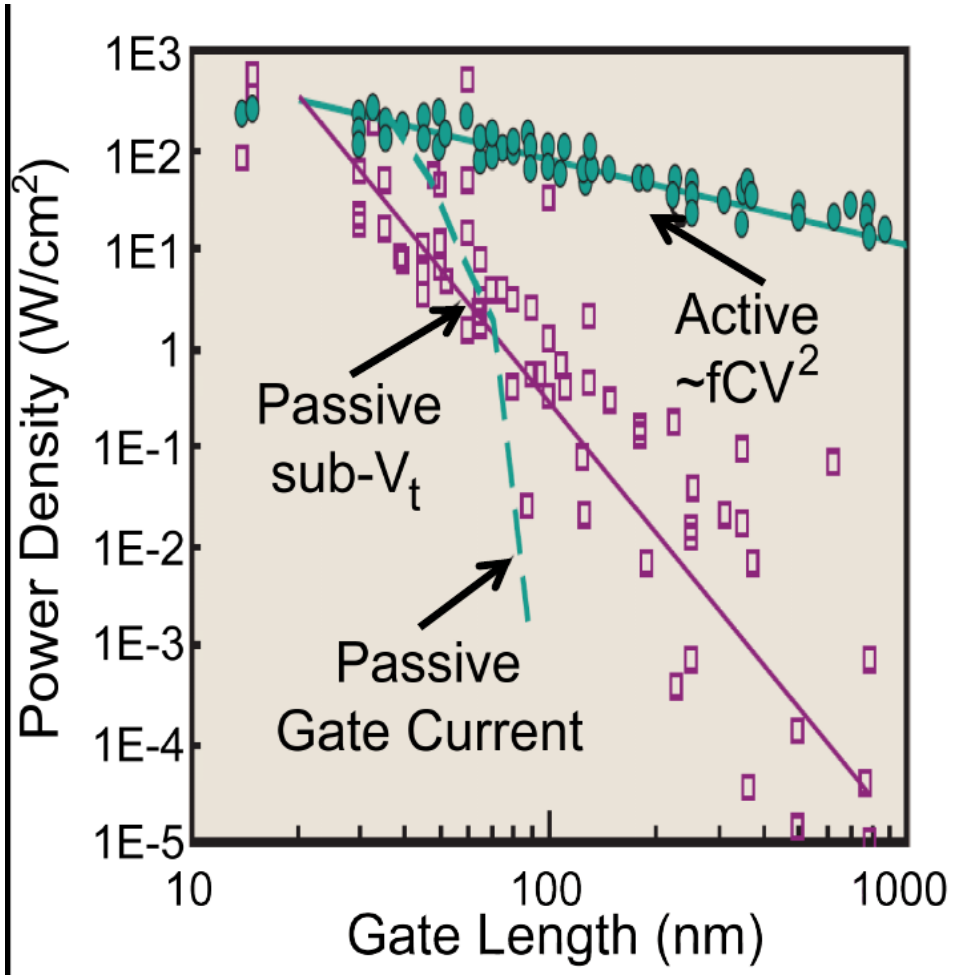


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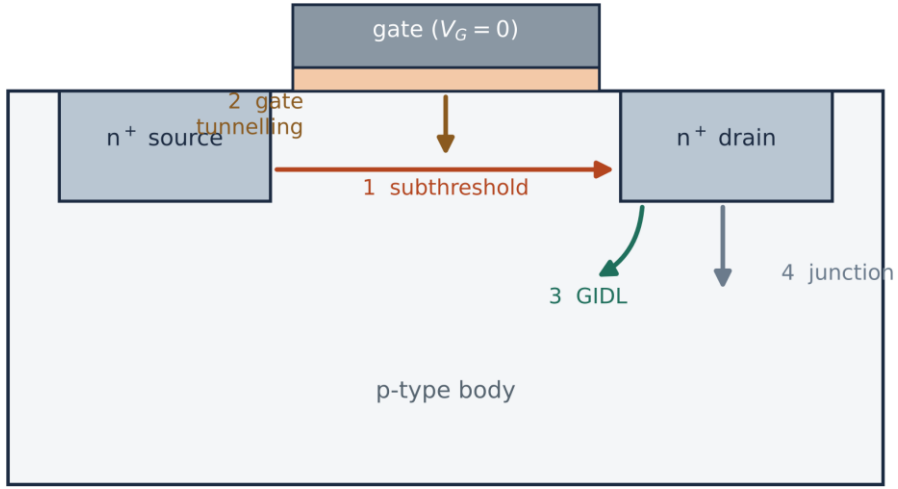
Symptoms: leakage

10^{10} devices × a few nA each = watts.

Leakage: from one device to the whole chip



Active and passive power density against gate length. W. Haensch et al., IBM J. Res. Dev. 50, 339 (2006); reproduced from ETH Zürich lecture slides.



The four leakage paths in an off-state n-MOSFET.

Four leakage paths

1

Subthreshold

Dominant. Set by V_T and SS, so worsened by roll-off, DIBL and swing degradation **together**.

2

Gate tunnelling

Exponential in physical thickness: why SiO_2 failed below ~1.2 nm and **high-k** arrived.

3

GIDL

Gate-drain overlap field drives **band-to-band tunnelling**. Sets a floor that V_T engineering cannot get below.

4

Junction

Reverse-biased drain–body leakage, with a band-to-band component at today's doping.

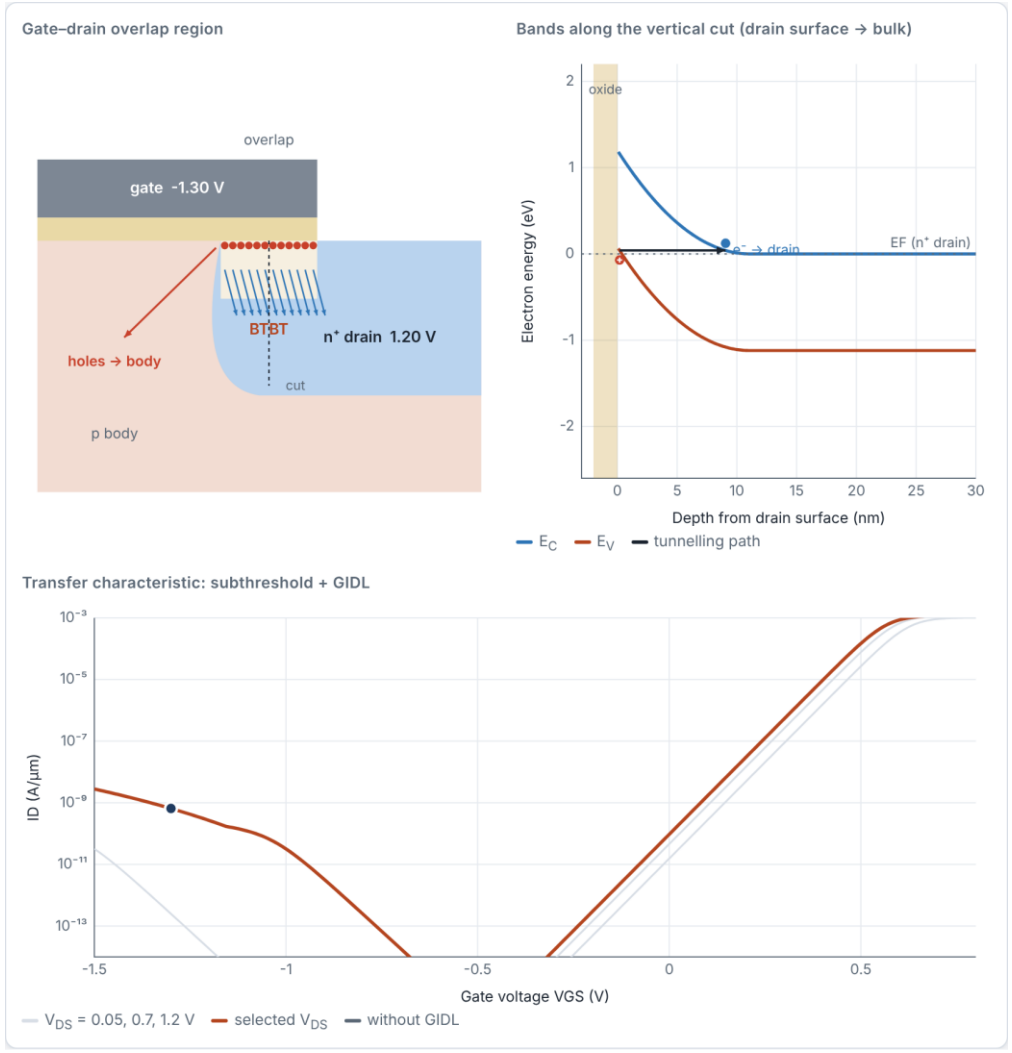
Animation: gate-induced drain leakage (GIDL)

Interactive animation 10

Gate-induced drain leakage (GIDL)

faizverse.github.io/nanoelectronic-devices/L1/gidl.html

- V_{DS} = 1.2 V: take V_{GS} from +0.3 V to −1.5 V and watch the bands bend past the gap
- Find the minimum of the transfer curve
- Thin the oxide: the GIDL tail moves toward 0 V
- Often the limiting leakage in low-power and DRAM technologies



The engineering consequence

- Static power is per-device leakage **times device count times supply**
- Static power became comparable to switching power in the **early 2000s**
- V_T scaling stopped, and **Dennard scaling ended** (Part 1, Section 4)
- One chain from 2-D electrostatics in one device to the architecture of every computer since

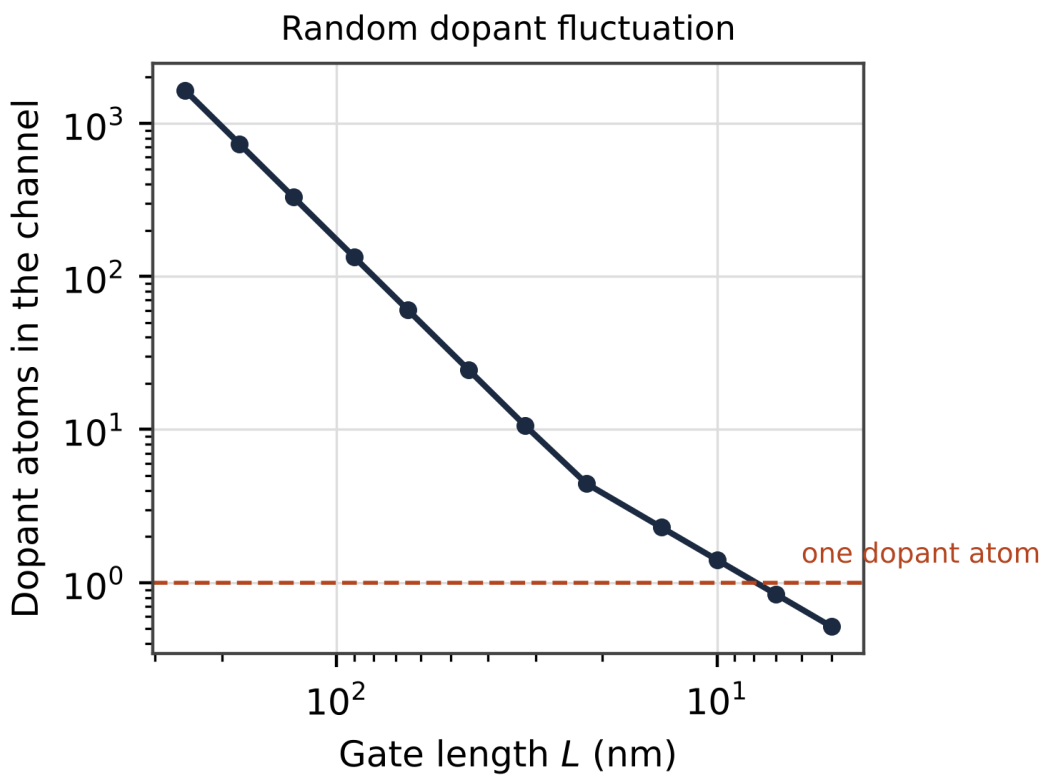
$$P_{static} = I_{OFF} \times N_{transistors} \times V_{DD}$$

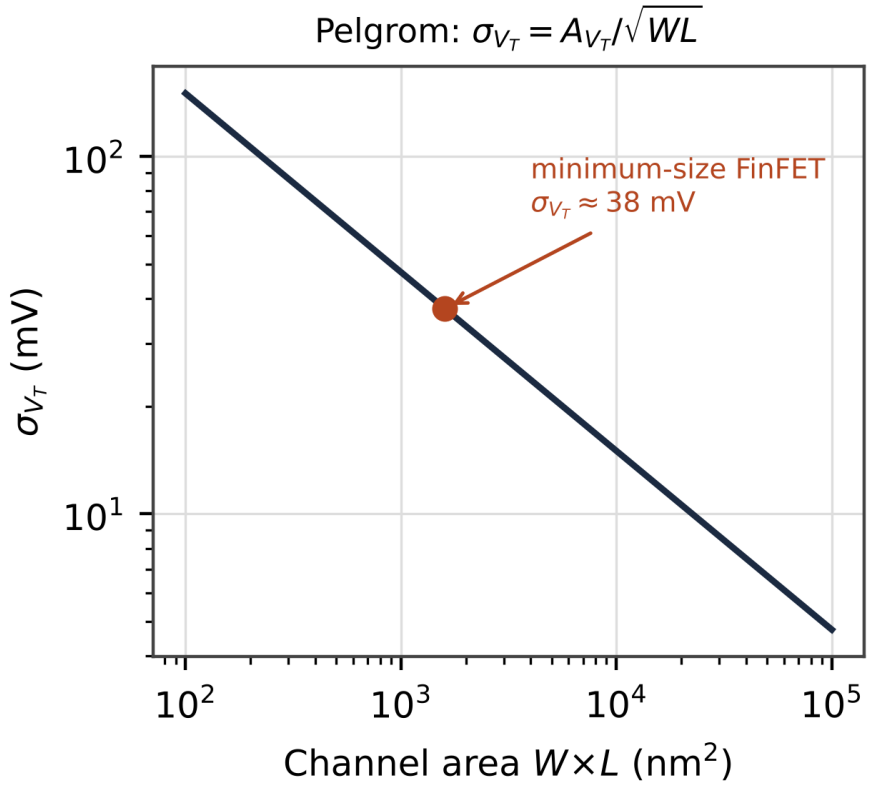
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Symptoms: variability

You no longer design the nominal device; you design the tail of the distribution.

Counting atoms





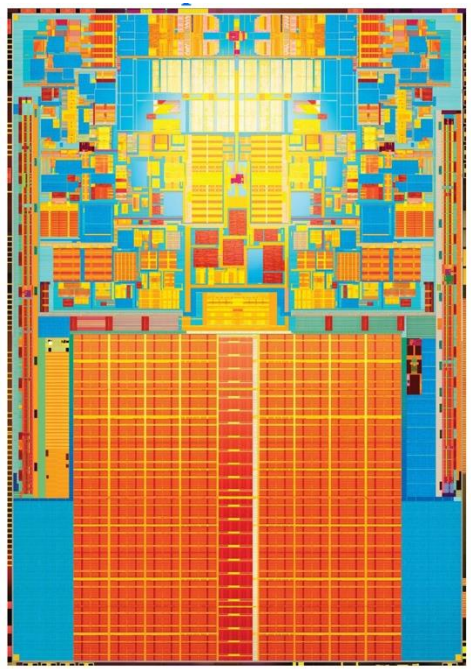
Left: dopant atoms in the channel volume as L shrinks (W = L). Right: Pelgrom scaling, illustrative A_{VT} = 1.5 mV·µm.

A 15 × 15 × 20 nm channel at 10^{18} cm^{-3} holds about **five dopant atoms**. Not five million: five.

Sources of variability

- **Random dopant fluctuation:** Poisson numbers and random positions; the deepest reason for **undoped** channels in FD-SOI, FinFET and GAA
- **Line-edge roughness:** 1–2 nm on a 12 nm gate; steep roll-off turns it into V_T spread
- **Metal-gate granularity:** a few grains with different work functions
- **SRAM** is the worst case and often sets the chip's minimum voltage. With 10^{10} devices, a one-in-a-million outlier occurs ten thousand times

$$\sigma_{V_T} \approx \frac{q}{C_{ox}} \sqrt{\frac{N_A W_{dm}}{3\,WL}} \propto \frac{1}{\sqrt{WL}}$$



Intel 45 nm die (2008): regular arrays are SRAM. Image: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

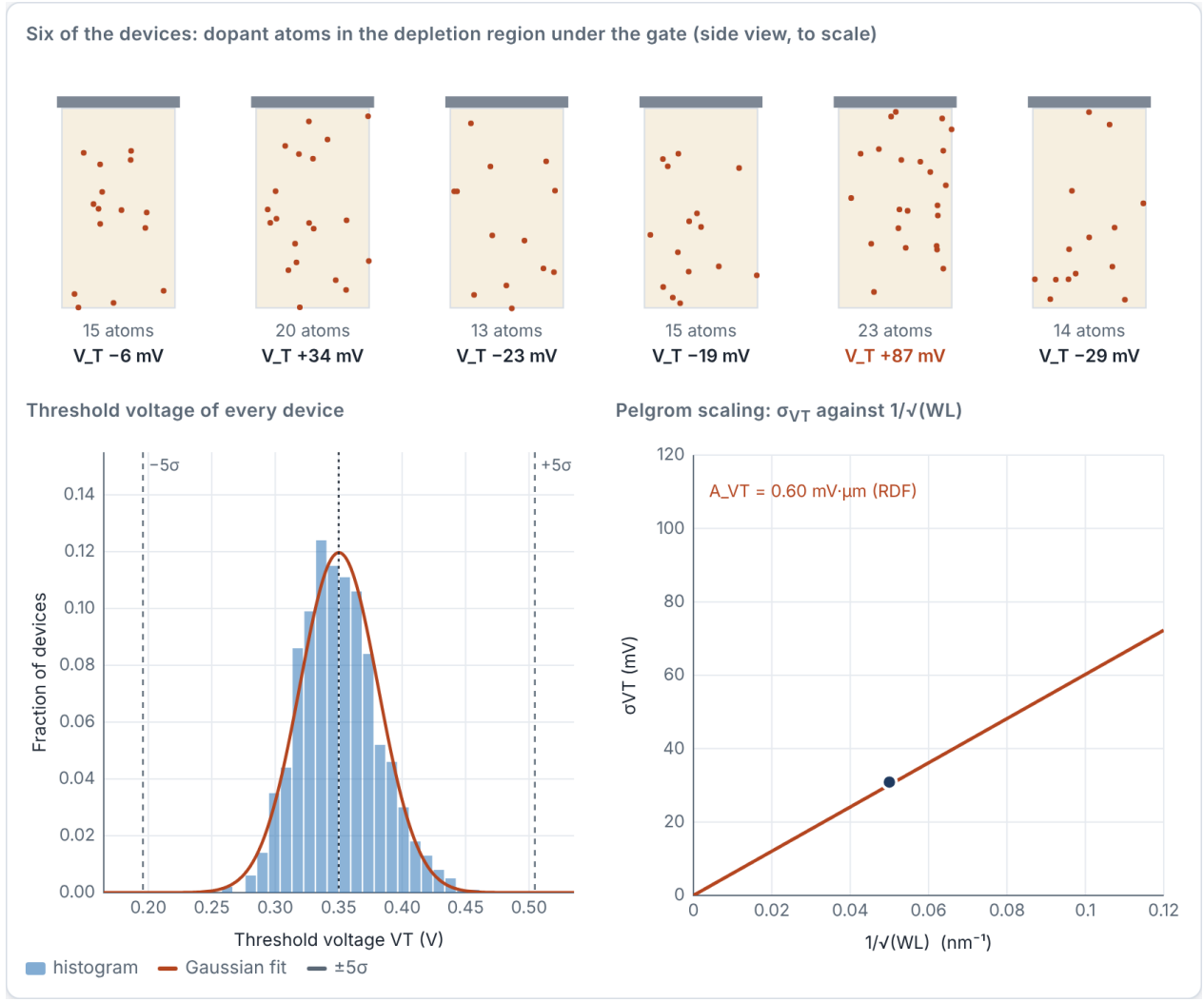
Animation: variability: random dopants

Interactive animation 11

Variability: random dopants

faizverse.github.io/nanoelectronic-devices/L1/variability.html

- W = L = 200 nm: thousands of atoms, a narrow histogram
- Shrink to 20 × 20 nm: count the atoms and read σ_{VT}
- Look at the ±5σ lines: does a 6-T SRAM cell still work?
- Undoped channel: RDF vanishes; other sources remain

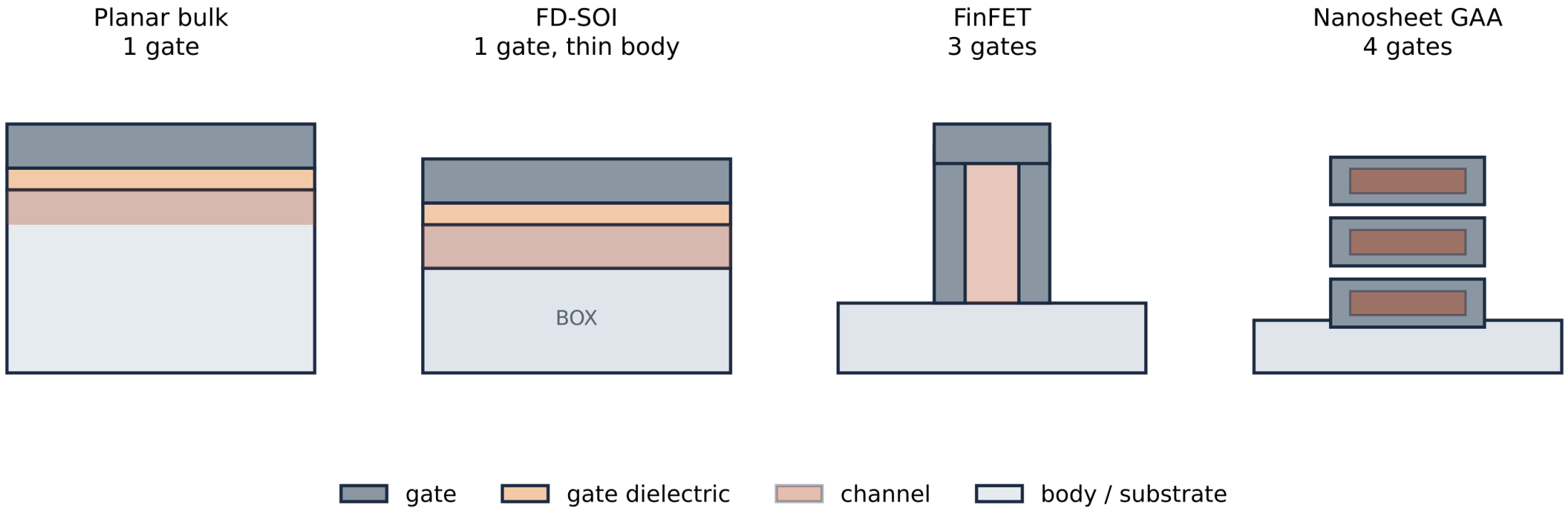


6

One family of solutions

Four things you can do to one equation.

Shrink the volume the drain can reach



Each step cuts the semiconductor volume the drain field reaches through, and adds sides from which the gate screens it.

Four levers on one equation

Together they take **L_{min} from 39 nm to about 10 nm**. That table is the roadmap of this course.

1

Reduce t_{ox}

High-k / metal gate (2007). A square-root lever, now exhausted: EOT ≈ 0.8–1 nm plus 0.3–0.4 nm quantum penalty.

2

Reduce x_j, add halos

Carried bulk planar to 28/20 nm. Costs: more GIDL, reverse short-channel effect, and heavy doping that **maximises RDF**.

3

Reduce t_{Si}

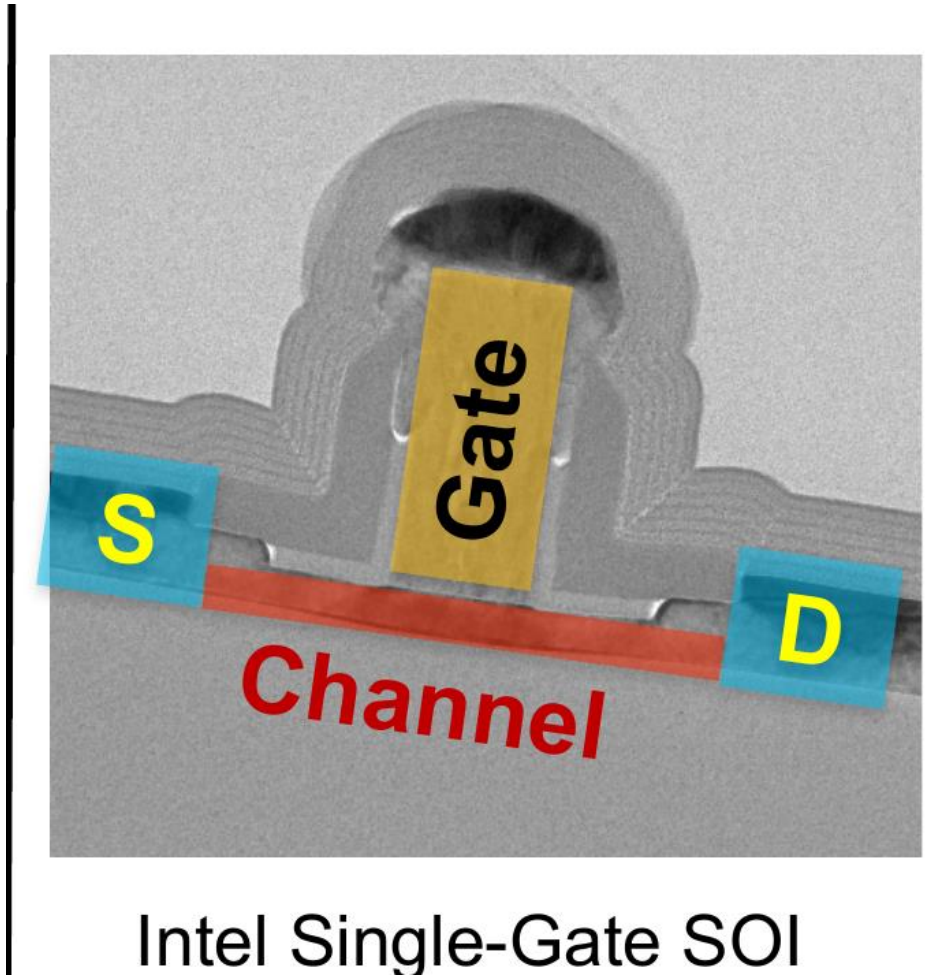
Fully depleted SOI: no 'beneath' for punchthrough; undoped body. Still in production at 28/22 nm for low power and RF.

4

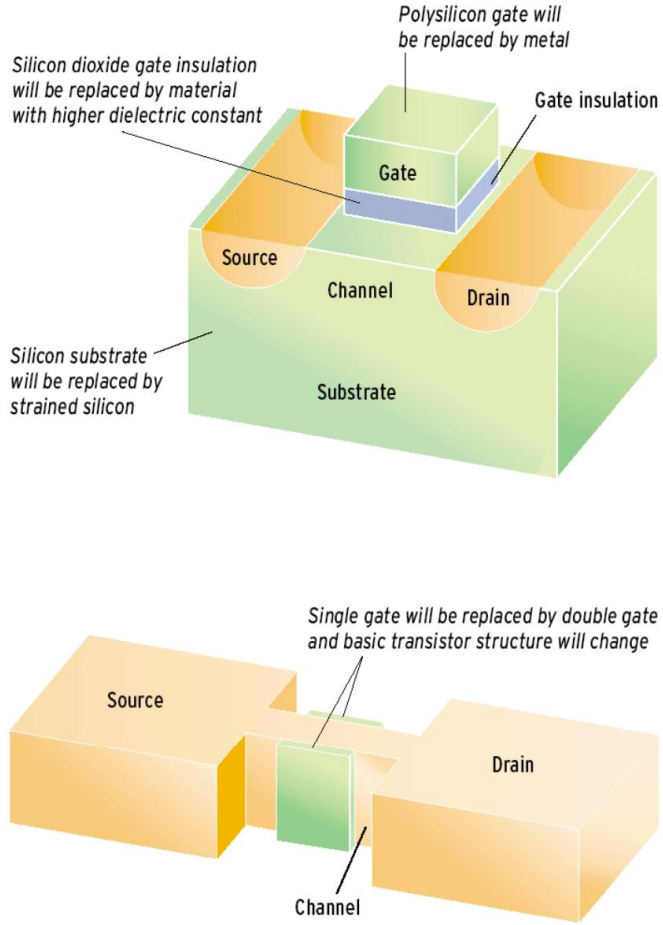
Add gates

FinFET (≈ 3), GAA (≈ 4) divide λ^2 by the gate count. **The strongest lever**: both architecture changes in Part 1.

Thin bodies and double gates



Intel experimental single-gate SOI transistor. R. Chau et al., DRC 2003; reproduced from ETH Zürich lecture slides.



A 2002 prediction: high-k, metal gate, strained Si, double gate. All four happened. IEEE Spectrum, Oct 2002; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011), after O. Brandt.

When the levers run out

Stack the devices

CFET (Part 1, Section 5.6): buys density, not a better λ.

Change the channel

Lower effective mass, higher injection velocity: **III–V, carbon nanotubes** (Lecture 4), 2-D materials such as MoS_2.

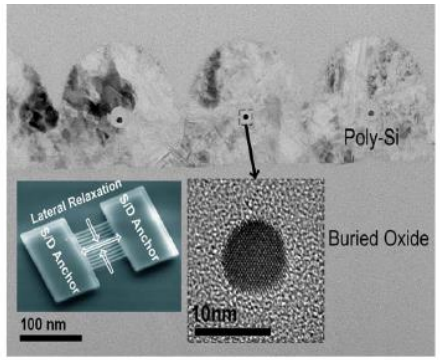
Change the injection

Tunnel FETs replace thermionic emission with band-to-band tunnelling (physics in Lecture 3).

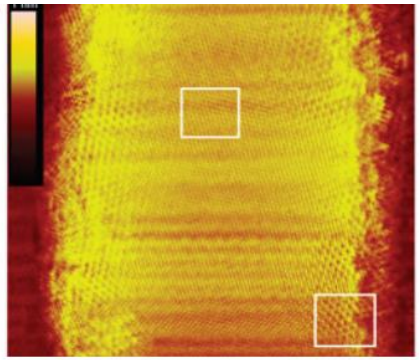
Why the module goes on

Perfect electrostatics still leaves SS at **60 mV/dec**: a thermodynamic floor. That is why the course continues past CMOS.

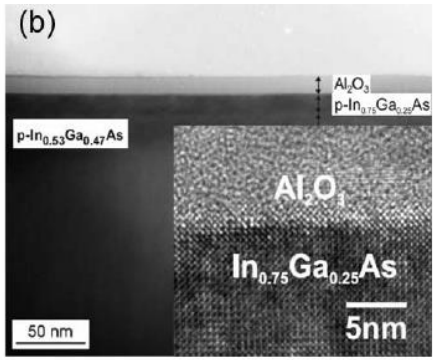
Post-silicon candidates, as seen around 2009



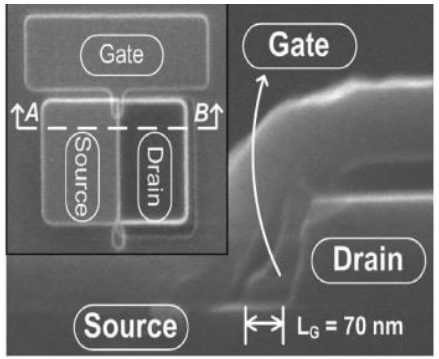
Nanowire



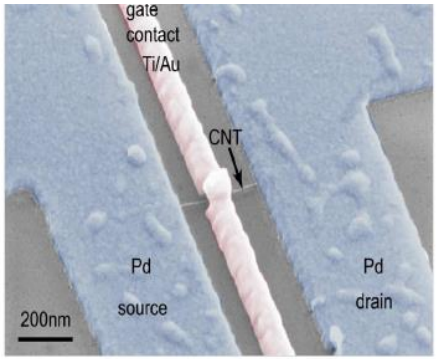
Graphene



III-V UTB



BTB Tunneling



CNT

Nanowire, graphene, III–V ultra-thin body, tunnelling FET, carbon nanotube. Sources: Hashemi et al., IEEE EDL 2009; Tapasztó et al., Nat. Nanotechnol. 2008; Wu et al., IEEE EDL 2009; Choi et al., IEEE EDL 2007; S. Guha, IBM; reproduced from ETH Zürich lecture slides.

The nanowire became today's nanosheet; the others remain research, and the subject of Lectures 2–4.

Summary: one idea, one equation

Group	Effects
Loss of gate control (Section 2)	V_T roll-off, DIBL, punchthrough, SS degradation, CLM
High fields (Section 3)	Velocity saturation, mobility degradation, impact ionisation, hot carriers
Leakage (Section 4)	Subthreshold, gate tunnelling, GIDL, junction
Variability (Section 5)	RDF, LER/LWR, metal-gate granularity

The one idea: the gate must control the source-to-drain barrier; short channels let the drain compete. **The fix:** four levers on λ, taking L_{min} from 39 nm to ~10 nm. **The limit beyond them:** 60 mV/dec, thermodynamic. Escaping it needs new injection or new materials: Lectures 2, 3 and 4.

Figure credits

Reproduced figures

ETH Zürich lecture slides: quantum-transport I–V (13–4 nm); power density (W. Haensch et al., IBM J. Res. Dev. 50, 339, 2006); SOI transistor (R. Chau et al., DRC 2003); post-silicon candidates (Hashemi, Tapasztó, Wu, Choi, Guha). W. A. Doolittle, Georgia Tech lecture slides (2011): Intel 45 nm die photo; IEEE Spectrum (Oct 2002) double-gate prediction, after O. Brandt. Copyright remains with the original owners.

Made for this course

Roll-off, barrier, charge-sharing, subthreshold, leakage, variability and architecture figures were generated for KKKC6353. Equations typeset for these slides. Animation screenshots from faizverse.github.io/nanoelectronic-devices.